QQ-A-250/23A
March 16, 1972
SUPERSEDING
Int. Fed. Spec. QQ-A-00250/23(ASG)
February 17, 1969

FEDERAL SPECIFICATION

ALUMINUM ALLOY ALCLAD ONE SIDE 7079, SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy alclad one side 7079 sheet; the general requirements are covered in QQ-A-250. The sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7079) core with a thin layer of an aluminum alloy (7072) anodic to the core bonded to one surface.

1.2 Classification.

1.2.1 Tempers. The sheet shall be classified as 0, T6, T62, or F temper, as specified (see 6.2). The definitions of these tempers shall be as specified in ANSI Std. H35.1.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in

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the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standard:

H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Core and cladding. The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the alclad sheets shall conform to the requirements specified in table I for core and cladding, respectively.

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TABLE I. Chemical composition 1/

Element	Analysis			
	Core alloy (7079)		Cladding alloy (7072)	
	Percent		Percent	
	Minimum	Maximum	Minimum	Maximum
Zinc	3.8	4.8	0.8	1.3
Magnesium	2.9	3.7	-	0.10
Copper	0.40	0.8	-	.10
Chromium	.10	.25	-	-
Manganese	.10	.30	-	.10
Iron	-	.40	-	2/
Silicon	-	.30	-	2/
Titanium	-	.10	-	-
Others, each	-	.05	-	.05
Others, total	-	.15	-	.15
Aluminum	Remainder		Remainder	

1/ Analysis shall be made regularly only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

2/ Iron plus silicon: 0.7 percent maximum.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties of the material perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset, minimum	Elongation in 2 in. or 4 times D 1/, 2/, minimum
	Inches	P.s.i.	P.s.i.	Percent
O	0.020 thru 0.062	3/38,000	3/21,000	10
	.063 thru .249	3/39,000	3/21,000	10
T6	.020 thru .039	69,000	60,000	7
	.040 thru .062	71,000	62,000	8
	.063 thru .249	72,000	63,000	8
T62 4/	.020 thru .039	68,000	58,000	7
	.040 thru .062	69,000	60,000	8
	.063 thru .187	70,000	60,000	8
	.188 thru .249	71,000	61,000	8
F	.020 thru .249	5/	5/	5/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ Material in the T62 temper is not available from the materials producers.
5/ No requirements.

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3.2.2 Mechanical properties after heat treatment. In addition to conforming to 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers shall, after proper solution heat treatment and aging, also conform to the requirements of table II for the T62 temper. Material as received in the T6 temper shall, after proper re-solution heat treatment and aging, be capable of conforming to the requirements specified in table II for the T62 temper.

3.2.3 Bend test. Bend-test specimens taken from material, shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

TABLE III. Bend test factor "N"

Thickness	Tempers	
	O	T6 and T62
Inch		
0.020 thru 0.032	1	7
.033 thru .063	2	8
.064 thru .091	3	9
.092 thru .125	4	10
.126 thru .249	5	11

3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7072) plates that are bonded to one side of the aluminum alloy (7079) ingot or slab to form a composite that is to be rolled to the finished thickness, shall have a thickness as specified in table IV.

TABLE IV. Cladding thickness

Thickness of finished sheet	Average thickness of cladding on the clad side of the finished sheet; minimum percent of sheet thickness
Inches	
0.020 thru 0.062	3.2
.063 thru .187	2
.188 thru .249	1.2

3.3.2 Thickness of cladding. If question arises concerning the thickness of cladding of the finished sheet, samples examined in accordance with QQ-A-250 shall show an average of cladding on the clad side, not less than that specified in table IV.

3.5 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 temper shall be identified by a lot number marked in at least one location on each piece.

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4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250)

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-250)

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in high-strength aeronautical applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).
(c) Dimensions required.
(d) Requirements for sizes not specifically covered (see QQ-A-250).
(e) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

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